

Severe Drug-Induced Autoimmune Hepatitis Triggered by Cardioril Therapy: A Well-Documented Idiosyncratic Liver Injury Case

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ABSTRACT -- Background: Cardioril (cardioril hydrochloride) is a novel vascular smooth muscle modulator indicated for refractory essential hypertension. Although pre-marketing clinical trials reported mild transaminitis (<1.2%), post-marketing pharmacovigilance is vital to characterize severe idiosyncratic autoimmune toxicities.

Case Presentation: A 61-year-old Caucasian male with a 10-year history of refractory hypertension and hyperlipidemia developed acute jaundice, scleral icterus, dark urine, and severe fatigue 42 days (6 weeks) following initiation of Cardioril 40 mg PO QD. Marked hepatic transaminase elevations (ALT 680 U/L [>12x ULN], AST 510 U/L [>12x ULN], Total Bilirubin 6.2 mg/dL, Alkaline Phosphatase 240 U/L) and antinuclear antibody seroconversion (ANA titer 1:640, speckled pattern; IgG 2,100 mg/dL) were confirmed. Percutaneous core liver biopsy demonstrated severe interface hepatitis with dense portal lymphoplasmacytic infiltrate and bridging necrosis without cirrhosis.

Management & Outcome: Inpatient admission was required. Immediate cessation of Cardioril (positive dechallenge) coupled with a 4-week tapering course of oral prednisone resulted in complete clinical and biochemical resolution over 6 weeks (ALT 28 U/L, Total Bilirubin 0.8 mg/dL). Updated RUCAM causality score was 8 ("Probable").

Keywords: Cardioril (cardioril hydrochloride); Drug-Induced Liver Injury (DILI); Drug-Induced Autoimmune Hepatitis; Scleral Icterus; RUCAM; Inpatient Hospitalization; Pharmacovigilance.

1. INTRODUCTION

Idiosyncratic drug-induced liver injury (DILI) remains a formidable challenge in clinical practice and drug safety. Among DILI phenotypes, drug-induced autoimmune-like hepatitis (DI-ALH) is particularly treacherous, often masquerading as idiopathic autoimmune hepatitis with elevated circulating autoantibodies and hypergammaglobulinemia [1]. Cardioril (cardioril hydrochloride) is a recently approved third-generation selective vascular smooth muscle receptor antagonist approved for refractory essential hypertension. In phase III trials, transient, asymptomatic alanine aminotransferase (ALT) elevations were documented in 1.1% of subjects; however, no cases of severe immune-mediated hepatitis or jaundice were reported prior to regulatory licensure. Here, we describe an unambiguous, biopsy-proven case of severe acute DI-ALH following Cardioril monotherapy in an adult male, establishing crucial baseline safety signals for pharmacovigilance surveillance.

2. PATIENT HISTORY & PRESENTATION

A 61-year-old Caucasian male presented to our emergency department with a 5-day progressive history of extreme fatigue, malaise, anorexia, tea-colored dark urine, and painless jaundice with scleral icterus. His past medical history was notable for a 10-year history of refractory hypertension and mixed hyperlipidemia. His outpatient maintenance regimen had included atorvastatin 20 mg PO daily for four years without hepatic dysfunction. Due to persistently elevated ambulatory blood pressures (158/96 mmHg), his primary cardiologist initiated Cardioril 40 mg PO once daily (QD) exactly 42 days prior to presentation. Baseline hepatic function panel obtained two weeks prior to Cardioril initiation was entirely normal (ALT 22 U/L, AST 19 U/L, total bilirubin 0.6 mg/dL, alkaline phosphatase 78 U/L). The patient reported strict adherence to the prescribed 40 mg daily dose.

He denied recent alcohol consumption, acetaminophen ingestion, over-the-counter herbal supplements, foreign travel, or blood transfusions. On physical examination, the patient was alert and oriented. Vital signs: blood pressure 134/82 mmHg, heart rate 74 bpm, respiratory rate 16 breaths/min, temperature 37.1 C. Marked cutaneous jaundice and conspicuous bilateral scleral icterus were evident. Abdominal examination revealed mild tenderness to palpation in the right upper quadrant without peritoneal signs. There was no hepatosplenomegaly, stigmata of chronic liver disease (spider angiomas, palmar erythema), or peripheral edema. Neurological examination showed no asterixis.

3. DIAGNOSTIC INVESTIGATIONS & SEROLOGY

Initial blood work revealed dramatic hepatocellular injury: ALT 680 U/L (>12x the upper limit of normal [ULN], ref 10-55 U/L), AST 510 U/L (>12x ULN, ref 10-40 U/L), Total Bilirubin 6.2 mg/dL (ref 0.2-1.2 mg/dL), Direct Bilirubin 4.1 mg/dL, and Alkaline Phosphatase 240 U/L (ref 40-130 U/L). The calculated R-value was 8.3, confirming a predominantly hepatocellular injury pattern. Serum albumin was 3.8 g/dL and INR was 1.15. A comprehensive infectious etiology workup was entirely negative: acute viral hepatitis serologies (anti-HAV IgM, HBsAg, anti-HBc IgM, anti-HCV IgG with HCV RNA PCR, and anti-HEV IgM) were non-reactive. Serologies for cytomegalovirus (CMV IgM) and Epstein-Barr virus (EBV VCA IgM) were likewise negative. Right upper quadrant ultrasound demonstrated normal hepatic parenchymal echogenicity, patent hepatic and portal veins with normal directional flow, and no biliary ductal dilatation or cholelithiasis.

Immunological evaluation revealed striking serological abnormalities: antinuclear antibodies (ANA) were strongly positive at a high titer of 1:640 with a speckled immunofluorescence pattern. Serum quantitative immunoglobulin G (IgG) was markedly elevated at 2,100 mg/dL (ref 700-1,600 mg/dL). Anti-smooth muscle antibodies (ASMA), anti-liver kidney microsomal type 1 (anti-LKM-1), and antimitochondrial antibodies (AMA) were negative.

Table 1. Serial Liver Biochemistries and Immunological Markers

Parameter	Reference	Baseline (Day -14)	Day +42 (Admission)	Day +84 (Resolution)
ALT (U/L)	10-55	22	680 (>12x)	28
AST (U/L)	10-40	19	510 (>12x)	24
Total Bilirubin (mg/dL)	0.2-1.2	0.6	6.2	0.8
Alkaline Phosphatase (U/L)	40-130	78	240	82
Serum IgG (mg/dL)	700-1600	--	2,100	1,180
ANA Titer (HEp-2)	<1:40	Negative	1:640	1:80

4. HISTOPATHOLOGY & CLINICAL COURSE

Given marked hyperbilirubinemia, severe transaminitis fulfilling Hy's Law criteria, and serological autoimmunity, inpatient hospitalization was initiated. A percutaneous core liver biopsy performed on hospital day 2 demonstrated *severe interface hepatitis with dense portal lymphoplasmacytic infiltrate and bridging necrosis*. Eosinophilic infiltration was noted in the portal tracts without established fibrosis (Ishak stage 0). These histopathological findings are classic for drug-induced autoimmune-like hepatitis and distinguish this entity from classic viral or alcoholic injury.

Cardioril was discontinued immediately upon admission on hospital day 1 (positive dechallenge). Supportive care and a planned 4-week tapering course of oral prednisone (initial dose 40 mg daily, tapered weekly by 10 mg: 30, 20, 10 mg) was instituted. The patient was discharged after four days of uncomplicated inpatient monitoring. Over the subsequent 6 weeks following discharge, serum transaminases and bilirubin steadily normalized (Table 1). At follow-up on day +84 (6 weeks post-discharge), the patient was completely asymptomatic with normal transaminases (ALT 28 U/L, AST 24 U/L, Total Bilirubin 0.8 mg/dL) and receding ANA titer (1:80). Rechallenge with Cardioril was considered strictly contraindicated and was not attempted due to clinical severity.

5. CAUSALITY ASSESSMENT & PV IMPLICATIONS

Application of the Roussel Uclaf Causality Assessment Method (RUCAM) for hepatocellular injury yielded an aggregate score of 8, signifying a "**Probable**" adverse reaction to Cardioril. Factors supporting causality include: (i) latency of 42 days (highly compatible with drug-induced autoimmunity); (ii) prompt dechallenge response with >50% normalization of ALT within 30 days; (iii) rigorous exclusion of alternative viral, biliary, and toxic etiologies; and (iv) biopsy demonstration of severe lymphoplasmacytic interface hepatitis with bridging necrosis [2,3]. The Simplified International Autoimmune Hepatitis Group (IAIHG) score was 6 ("Probable AIH"). Under regulatory pharmacovigilance criteria (ICH E2D, FDA 21 CFR 314.80), this event represents a **serious individual case safety report (ICSR)** meeting inpatient hospitalization and medically significant criteria. Clinicians should maintain vigilant baseline and monthly hepatic monitoring throughout the first 90 days of Cardioril therapy.

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